Change 1:-

→ Our requirements (earlier)

P_ℓ → Prob. of realizing ℓ^{th} RIS state

$\hat{P}_\ell$ → Emp. freq of " " " ".

1

Let $p \triangleq [P_1\ P_2 \cdots P_L]^T$

$\hat{P}(T_c) \triangleq [\hat{P}_1(T_c), \ldots, \hat{P}(T_c)]^T$

Requirement:-

$$\|\underline{p} - \underline{\hat{p}}(T_c)\|_{TV} < \varepsilon_1 \longrightarrow$$ This holds even now!

2 Let q_K be the prob. that UE-k is scheduled in T_c slots. But from Theorem 1, it is clear that UE-k will be scheduled in a slot only when RIS assumes a state which is in beamforming config. to the UE. Therefore in order for each UE's throughput to converge to its average long-term throughput, every UE must be scheduled sufficiently many times whenever the RIS assumes BF state to it.

For k^{th} UE, let $\ell^*(k)$ denote the index of RIS state that gives max-rate @ UE-k. Let $\mathcal{T}_k(T_c)$ denote the time slot indices within T_c slots where RIS assumes state $\ell^*(k)$ & $S_k(t)$ be UE-Scheduling indicator:-

$$\hat{q}_k(T_c) \triangleq \frac{1}{|\mathcal{T}_k(T_c)|} \sum_{t \in \mathcal{T}_k(T_c)} S_k(t).$$

$$q_k \triangleq \mathbb{E}\left[\hat{q}_k(T_c)\right].$$

<u>Note:-</u> Earlier q_k was unconditioned! Now q_k is conditioned on the BF-RIS configuration.

Suggested change!

2) **Convergence of per-UE scheduling frequency:** In order for every UE to achieve its own long-term average throughput under PF scheduling, each UE must be scheduled sufficiently many times over the window of T_c slots. From Theorem 1, we infer that a UE is scheduled only when the realized RIS configuration directs its reflection towards the angular region containing the UE. To this end, let $S_k(t) \in 0, 1$ denote the scheduling indicator for UE-k at time t, and let $\mathcal{T}_k$ denote the set of time slots for which the RIS state lies in the BF/angular region corresponding to UE-k. Then, we define the empirical scheduling frequency of UE-k, conditioned on the RIS configuration being in its corresponding BF state, as

$$\hat{q}_k(T_c) = \frac{1}{|\mathcal{T}_k|} \sum_{t \in \mathcal{T}_k} S_k(t), \tag{17}$$

and the empirical scheduling frequency vector: $\hat{\mathbf{q}}(T_c) \triangleq [\hat{q}_1(T_c), \ldots, \hat{q}_K(T_c)]^T$. Further, let $q_k = \mathbb{E}[\hat{q}_k(T_c)]$, and the corresponding expected scheduling vector be $\mathbf{q} = [q_1, \ldots, q_K]^T$. Then, it suffices to choose T_c such that the $\hat{\mathbf{q}}(T_c)$ is close to $\mathbf{q}$ with high probability.

We can formally capture the above requirements using the *total variation (TV) distance* between two probability

<u>Theorem 2:-</u>

<u>Change of bound:-</u>

Notice the change!

$$T_c \geq \max\left\{ T_s \frac{1}{2\epsilon_1^2} \ln\left(\frac{2K}{\eta_1}\right), \frac{1}{2(p_{min} - 2\epsilon_1)\epsilon_2^2} \ln\left(\frac{2K}{\eta_2}\right) \right\}$$

<u>Proof:-</u> Proceed similar to current proof!

$$\hat{q}_k = \frac{X_k}{N_k} \quad \text{where} \quad X_k = \sum_{t \in \mathcal{T}_k(T_c)} S_k(t)$$

$$N_k = |\mathcal{T}_k(T_c)|.$$

By using bounded-difference argument as you did now, you can similar arrive at:-

$$\Pr\left(|\hat{q}_k - q_k| \geq \epsilon_2\right) \leq 2\exp\left(-2N_k\epsilon_2^2\right)$$

$$N_k = T_c\,\hat{p}_{\ell^*(k)}$$

But we impose condition that $\|p - \hat{p}\| < \epsilon_1$

$$\Rightarrow |\hat{p}_\ell - p_\ell| \leq 2\epsilon_1$$

$$\therefore N_k \geq T_c\left(p_{\ell^*(k)} - 2\epsilon_1\right) \longrightarrow \text{A}$$

$$\Rightarrow \Pr\left(|\hat{q}_k - q_k| \geq \epsilon_2\right) \leq 2\exp\left(-2T_c\left(p_{\ell^*(k)} - 2\epsilon_1\right)\epsilon_2^2\right)$$

In order to have a uniform bound across all UEs,

Choose $p_{min} \triangleq \min_{\ell \in [L]} p_\ell$

Then $\Pr\left(|\hat{q}_k - q_k| \geq \epsilon_2\right) \leq 2\exp\left(-2T_c\left(p_{min} - 2\epsilon_1\right)\epsilon_2^2\right)$

Using union-bound arguments like you did:-

finally, you get:-

$$T_c \geq \frac{1}{2\left(p_{min} - 2\epsilon_1\right)\epsilon_2^2} \ln\left(\frac{2K}{\eta_2}\right)$$

Proof of Throughput convergence:-

Empirical throughput of UE-k:-

$$\tilde{T}_k = \sum_{\ell=1}^{L} \underbrace{\hat{p}_\ell}_{\text{unconditional prob}} \overbrace{\hat{q}_{k|\ell}}^{\text{conditional prob.}} R_{k,\ell}$$

$$T_k^* = \frac{1}{K} R^{opt} = \underbrace{p_{\ell^*(k)} q_{k|\ell^*(k)}}_{=\frac{1}{K}} R_{k,\ell} = \sum_{\ell=1}^{L} p_\ell \underbrace{q_k}_{=\frac{1}{K} \text{ if } \ell = \ell^*(k).} R_{k,\ell}$$

$$\tilde{T}_k - T_k^* = \sum_{\ell=1}^{L} \left(\hat{p}_\ell \hat{q}_{k|\ell} - p_\ell q_{k|\ell}\right) R_{k,\ell}$$

Add & subtract $p_\ell \hat{q}_k$:-

$$\tilde{T}_k - T_k^* = \sum_{\ell=1}^{L} (\hat{p}_\ell - p_\ell)\hat{q}_k + p_\ell(\hat{q}_k - q_k) R_{k,\ell}$$

$$\Rightarrow \left|\tilde{T}_k - T_k^*\right| \leq \sum_{\ell=1}^{L} |\hat{p}_\ell - p_\ell| |\hat{q}_k| R_{k,\ell} + \sum_{\ell=1}^{L} |\hat{q}_k - q_k| p_\ell R_{k,\ell}$$

$$= \underbrace{|\hat{q}_k|}_{\leq 1} \underbrace{\sum_{\ell=1}^{L} |\hat{p}_\ell - p_\ell|}_{\leq 2\varepsilon_1} \underbrace{R_{k,\ell}}_{\leq R_k^{opt}} + \underbrace{|\hat{q}_k - q_k|}_{\leq 2\varepsilon_2} \underbrace{\sum_{\ell=1}^{L} p_\ell}_{=1} \underbrace{R_{k,\ell}}_{\leq R_k^{opt}}$$

$$\therefore \left|\tilde{T}_k - T_k^*\right| \leq (2\varepsilon_1 + 2\varepsilon_2) R_k^{opt}$$